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Statement of Purpose

I wish to create and contribute to systems that address resource management and scheduling, scalability in operating systems, memory and storage optimization, and the unification of heterogeneous architectures. As a Master's student at UC San Diego, I learned the importance of asking probing questions. This curiosity has driven me towards problems whose very existence I might not have been aware of otherwise.

During my first quarter at UC San Diego, I enrolled in Prof. Yuanyuan Zhou's Graduate Operating Systems course. This was a transformative experience for me as it diverged from the traditional approach of asserting "X is right and Y is wrong" and instead focused on understanding various authors' rationale behind choosing X over Y. I was intrigued when other authors opted Y rather than X, for a reasoning that best suited their respective systems. I came to realize that computer systems are quite philosophical, where absolute right or wrong is replaced by numerous tradeoffs.

My first hands-on research project was with Prof. Amy Ousterhout. This project focused on improving CPU efficiency by offloading datacenter tax - encryption, compression, and copying - to SmartNIC accelerators. To evaluate performance disparities, I identified applications that spent substantial CPU cycles on datacenter tax and conducted benchmarking analyses with and without offloading the tax. I started with the Synthetic Web Service application created by AIFM (Application-Integrated Far Memory). Since performance was to be benchmarked on a single machine, I isolated the application's data structures from inherent far memory usage and then migrated the Shenango-based application over to Caladan, a novel CPU scheduler that supplanted Shenango. After offloading compression tasks to a Mellanox ConnectX-5 SmartNIC, we saw significant performance improvements - specifically, a **3.5x boost in throughput**, and a **20% reduction in latency**. I found this work particularly intriguing due to its pragmatism: in today's datacenters, there is a persistent surge in server requirements, further intensified by the diminishing effects of Moore's Law. By offloading datacenter tax to hardware accelerators, there is a remarkable potential to enhance CPU efficiency and curtail costs associated with scaling computer systems.

Another project I worked on was Brim, a novel far memory system, with Prof. Amy Ousterhout, Prof. Alex Snoeren, and Anil Yelam, a PhD student at UCSD. Most far-memory systems today use kernel-based paging, incurring significant overhead and limiting application control over memory management. Brim incorporates a hint API to alert the pager about anticipated cold memory accesses, while preserving the conventional page-based memory interface. I contributed to its evaluation by comparing the performance of memory-intensive applications like DataFrame and AIFM's applications with Brim and alternative far memory systems. We observed Brim's performance to be up to **150% higher than Fastswap**, and this work is under review at a top-tier systems conference. This allowed me to explore the new area of far memory systems and comprehend why they exist in the first place. I also learned the immense value of collaboration and bringing together individuals from diverse disciplines to engineer a solution. The professors' strong emphasis on the

entire process of reproducing results highlighted the significance of ensuring accurate and reliable findings.

As an undergraduate junior, I conducted seminars to teach Computer Architecture and Python. I realize the responsibility of teaching others only after attaining a thorough understanding myself. As a summer fellow at IIT Bombay, I created comprehensive lecture notes and animations for various topics in Advanced Mathematics under the guidance of Prof. Prabhu Ramachandran. I learned the importance of citing precise references, ensuring that readers have access to relevant and reliable sources of information. These experiences gained from teaching and creating educational resources motivate me to pursue a PhD and I wish to make meaningful advancements to systems research.

When the pandemic hit, students had to repeat their programming assignment (PA) submissions on many disjoint platforms because of the limited capabilities of each. Ironically, the fundamental principle of programming, "Don't Repeat Yourself," was not upheld. To address this issue, I created a unified web environment for graders to execute PAs with an incorporated pseudoterminal, perform plagiarism checks, and grade submissions. I founded SudoLMS to commercialize this project, which garnered over 600 active users at its peak. This was when I first realized how creating novel solutions excites me.

Pursuing a PhD will allow me to create systems while completely rethinking the foundations over which current systems are built, and continue exploring new areas of systems research. The PhD program at UT Austin will allow me to explore these interests. I would especially like to work with Prof. Chris Rossbach, Prof. Emmett Witchel and Prof. Daehyeok Kim. Prof. Rossbach's work on machine learning for systems and unifying heterogeneous architectures aligns well and greatly overlaps with my interests. I would also like to work with Prof. Witchel at the intersection of storage systems, cloud computing and parallelism. Additionally, Prof. Kim's work on operating systems for datacenters and hardware accelerators deeply resonates with my experiences in these fields.

After completing graduate studies, I wish to pursue a career in academia as well as contribute to systems research at industry. I love to teach - the concept of conveying my understanding of ideas to someone, in a way that they can understand, truly makes me happy. A career in academia gives me this opportunity to teach and attempt to shape the academic journey of the next generation of bright individuals.